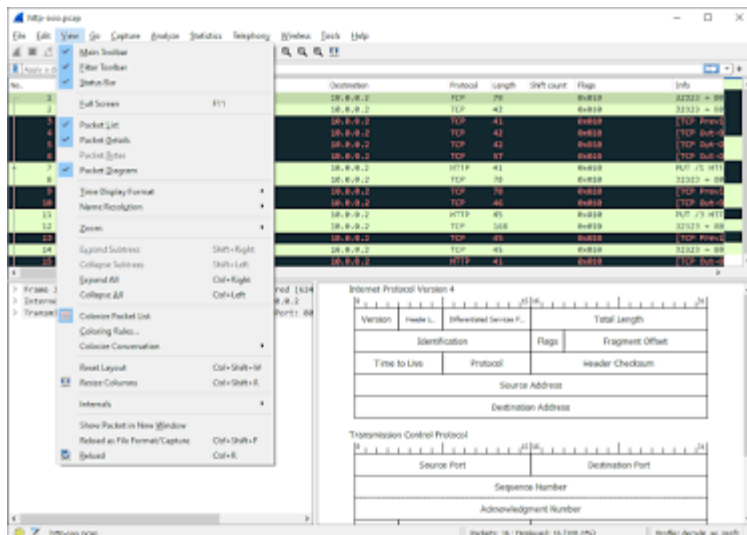


Day 1: Introduction to Network Protocols Theory

Task 1 :- Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.

ANS :-

Step 1: Record Traffic :-



Open Wireshark, select your network connection (Wi-Fi or Ethernet), and click the blue shark fin to start recording.

Open your browser and go to [www.spotify.com](<https://www.spotify.com>).

Stop the recording using the red square button.

Step 2: Find the Packets :-

Find 2 DNS Packets :-

- Type dns in the Wireshark filter bar and press Enter.
- Look for these two lines :
 1. DNS Query : Standard query ... A
www.spotify.com (Your PC asking "What is Spotify's IP address?")
 2. DNS Response : Standard query response ... A ... (The server replying with Spotify's IP address)

Find 2 HTTPS Packets :-

- Type tls (or http) in the filter bar and press Enter.
- Look for these two lines:
 1. Client Hello : Client Hello (Your PC introducing itself to secure the connection)
 2. Server Hello : Server Hello (Spotify's server replying to establish the secure connection)

Task 2 :- Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

ANS :-

```
nslookup www.flipkart.com
```

```
Server: 1.1.1.1
```

```
Address: 1.1.1.1
```

```
Non-authoritative answer:
```

```
Name: www.flipkart.com
```

```
Address: 163.53.76.86
```

What it means :

- **The IP Address (163.53.76.86) :** This is the actual numeric internet address of Flipkart's web server.
- **The Meaning :** Computers don't use web names; they use numbers. The DNS lookup acts like a phonebook—it converts the human-friendly name [www.flipkart.com](<https://www.flipkart.com>) into the numeric IP address so your browser knows where to connect.

Task 3 :- Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

ANS :-



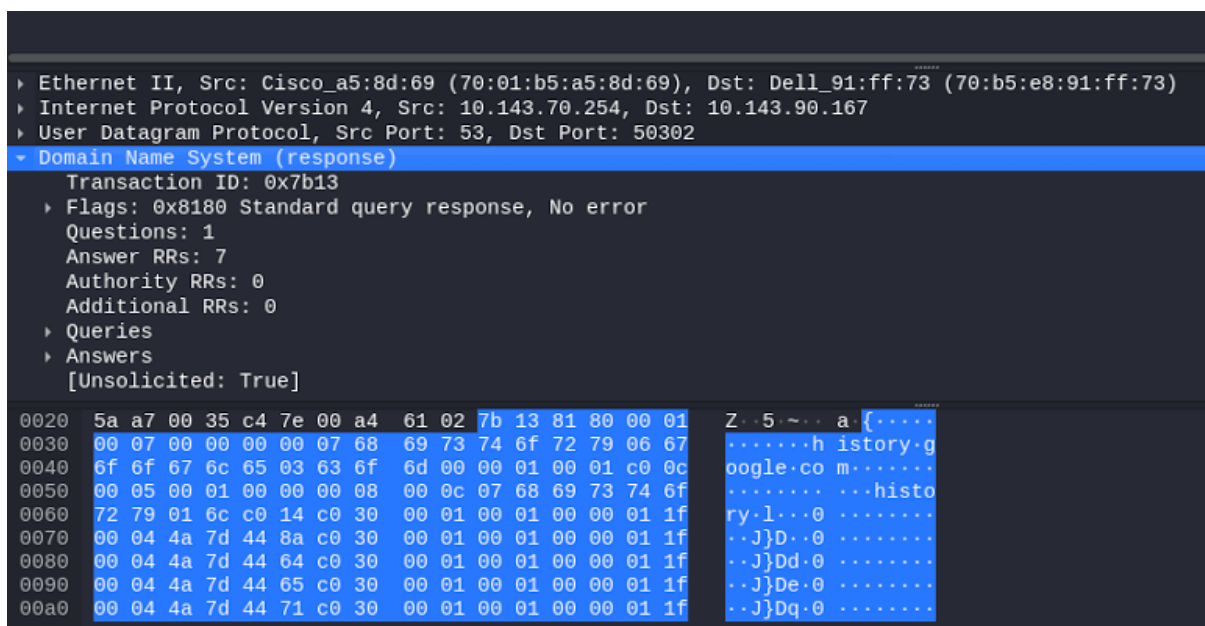
1. Security : HTTP sends data in plain text (anyone can read it). HTTPS encrypts your data (locks it so hackers can't read it).
2. Padlock Icon : HTTP displays "Not Secure" in your browser bar. HTTPS shows a padlock icon 🔒 next to the web address.
3. Port Number : HTTP uses port 80. HTTPS uses port 443.

Real-World Example :-

- Google Pay / PayPal / Online Banking Apps: These apps use HTTPS to ensure your bank details, passwords, and payment info are encrypted and safe while sent over the internet.

Task 4 :- Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.

ANS :-



Query 1 (IPv4 Request) :

- Message : Standard query ... A newwebsite.com
- Meaning : Your PC asks the DNS server, "What is the IPv4 address for this site?"

Query 2 (IPv6 Request) :

- Message : Standard query ... AAAA newwebsite.com
- Meaning : Your PC asks simultaneously, "Does this site have an IPv6 address?"

Response 1 (Server Reply - IPv4) :

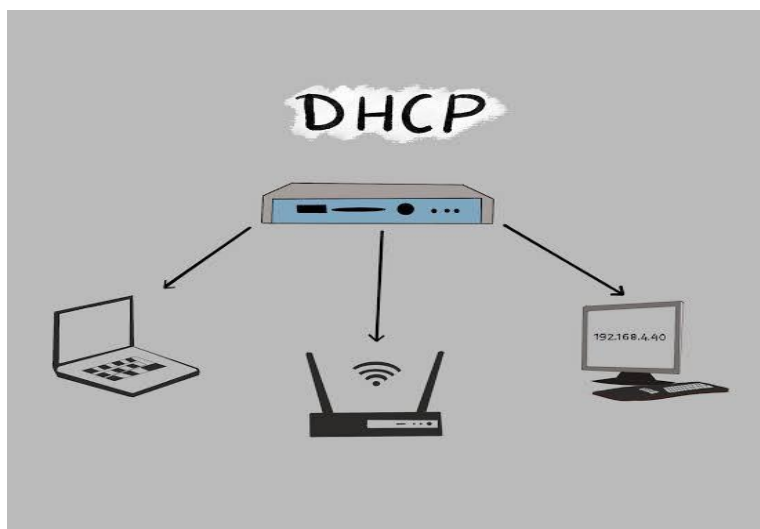
- Message : Standard query response ... A 192.0.2.1
- Meaning : The DNS server replies with the website's exact numeric IP address.

Response 2 (Server Reply - IPv6 / CNAME) :

- Message : Standard query response ... AAAA ... or CNAME ...
- Meaning : The server sends any alternative names or IPv6 addresses. Your browser now loads the site using the IP address it received.

Task 5 :- Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).

ANS :-



Scenario : Opening Instagram at a Café

1. **You Connect to Wi-Fi :** You walk into a café and tap "Café Wi-Fi" on your phone.
2. **Your Phone Asks (DHCP Request) :** Your phone shouts to the Wi-Fi router, *"I'm new here! Can I get an IP address to use the internet?"*
3. **The Router Assigns (DHCP Response) :** The café router automatically hands your phone a temporary address (like 192.168.1.15).
4. **Result :** Without DHCP, you would have to manually type complex network settings every time you join a new Wi-Fi network. Instead, DHCP hands you an IP address instantly so you can start scrolling Instagram right away.

